Promptly Done · An AI buyer for industrial procurement · Tacto Negotiation Arena

Numbers from code.
Words from the model.

The problem

Total cost.
Not price.

Payment days, coverage of the need, lead time, defects, tooling, index risk. Every term is euros - and the arena scores all of them.

Our buyer protects total cost, must-haves and supply continuity - and keeps the supplier relationship workable.

What we found in our own games

90%

of the value is decided by the supplier's reply to our first e-mail.

Reply 1: +0.46 score on average. Replies 2 to 4 together: +0.08.

The architecture

Code decides every number.

Data layer

Master JSON + SQL

Package, Sunday release, hidden-case facts in one store

→

Prepare

Dossier, claims, anchor

Buyer alternative, C of every term combination, floor - before the clock

→

Negotiate

One worker per thread

Analyst, supplier model, MESO steps, sheets, e-mail

→

Close

Accept ladder

Final offers, best sheet ever, wave cap

The model may profile a supplier, read its e-mail and phrase ours - with a deadline and a code fallback on every call. It never sets a price, a term or an accept.

The data pipeline

Arguments the arena can verify.

Sources

Contracts, orders, quotes, payments, quality, call notes, indices, cost drivers

→

One layer

Master JSON with the Sunday release and hidden-case facts, queried as SQL

→

Dossier

Per supplier: cash needs, capacity, scorecards, past settlements

→

Claims

Computed exactly as the arena checks them

Index moves: two months of one series, 12 months apart, in the latest window - only those that argue our side.

Notice periods: the letter's days against the contract's. P01: 51 given, 90 required.

Each data ID once per case. A wrong figure costs trust, so a claim that does not check out is never sent.

One negotiation round, per supplier thread

Read. Model. Target. Write.

1

Analyst

Its offer, refused terms, asks, evidence and tone - English and German. Every figure must appear in its text.

2

Supplier model

Full cost from its breakdown or should-cost / 1.02. Its cost of money from its own price pairs.

3

Target C

Anchor by category goal, then a shrinking concession curve: no free margin giveaways.

4

Sheets + e-mail

Up to three packages at the target. A refused term drops out; the e-mail pivots.

Every reply runs this loop inside the 60 seconds, measured per thread - the model gets only the time that is left.

The core technique · MESO

One term per e-mail.

Three packages that cost us the same, different in one term.

Its pick and its prices reveal what the term is worth to the supplier.

Lock it at our optimum. Vary the next. Price comes last.

Case P02 · payment days, priced by the supplier

Payment	Its price	Our total cost
30 days	369.02	2,269,744
60 days	371.08	2,267,299
90 days	373.14	2,264,686

Its money costs it 6.7 % a year, ours 8 %. Paying more for 90 days is the cheaper deal.

Rules from data · each seen in three cases or more

We read every game.

8/8

walked away when we wrote after a final offer. Now we accept at once.

30%

of replies hid an alternative sheet. We take the cheapest ever offered.

5.5x

better on P08 when every sheet covers the full need: 0.44 vs 0.08.

3-7x

its value is what suppliers charge for faster delivery. We ask only what we need.

~80 traces → **a written spec, R1 - R13** → **241 offline tests**

Recorded games replay through the code before anything ships.

The lab

The arena is deterministic.
So we ran experiments.

Knobs, not rewrites

Anchor depth, concession curve, reciprocity, close gap, number of term steps - 19 named variants.

A replay book

Identical moves replay from 250+ recorded games for free. Only a changed move costs a session.

A strict selection

A variant wins a case group only with +0.005 mean and no case lost by more than 0.05.

0.733 with e-mails written by code, vs 0.661 when the model wrote them. So in the final, code writes.

One story · case P02, copper +47.7 % in a year

0.507 → 0.735

A safety rule banned index clauses on volatile series. We removed it and let the cost function decide.

It already prices the expected index move and our risk. The supplier paid for the certainty - and the other nine cases replayed exactly as before.

Practice score · mean of 10 cases × 100

From 13 to 73 in one night.

13.3

First LLM agent. Pushed past final offers.

56.9

Close rules, full coverage, cost-capped sheets.

71.6

MESO steps, learned playbook, the lab.

73.3

The cost function decides on index clauses.

The live final · nothing may break

No human in the loop.

One worker per supplier

Each RFQ thread keeps its own 60-second clock.

Validated accepts

A refused offer moves to the next checked candidate - never to a must-have break.

Wi-Fi proof

Before a retry it checks whether the message already landed.

Closes before the cap

90 seconds before a wave's cap it takes the cheapest valid offer that beats the alternative.

Waits through the pre-wave 403, resumes a session after a 409, follows seconds_left. Never reads the leaderboard, event feed or screen. Rehearsed in a local two-wave simulator.

Let code count.
Let the supplier reveal.
Let data decide.

Thank you. Questions?